

INTRODUCTION TO QUBITS

State vectors, amplitudes, measurement, phase, Bloch-sphere geometry, basis changes, mixed states, and the bridge to multi-qubit systems

A detailed reference for quantum computing, quantum cryptography, and quantum information.

Topic 9 in the Quantum Mathematics Foundation series

How to use this volume

This chapter gives a self-contained description of the qubit as a mathematical and physical information unit. It assumes complex numbers, vectors, matrices, inner products, and elementary probability. The focus is not on memorizing gate tables, but on understanding how amplitudes, phases, basis choices, and measurement fit together.

Learning objectives

- Represent a pure one-qubit state as a normalized vector in $\mathbb{C}^2$.
- Convert amplitudes into measurement probabilities using the Born rule.
- Distinguish global phase from relative phase.
- Measure a state in Z, X, and Y eigenbases.
- Interpret pure states geometrically using the Bloch sphere.
- Explain single-qubit gates as unitary transformations and rotations.
- Use density matrices to distinguish pure states from classical mixtures.
- Understand how tensor products create multi-qubit state spaces and permit entanglement.

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1. Why qubits are different

A qubit is not merely an unknown classical bit

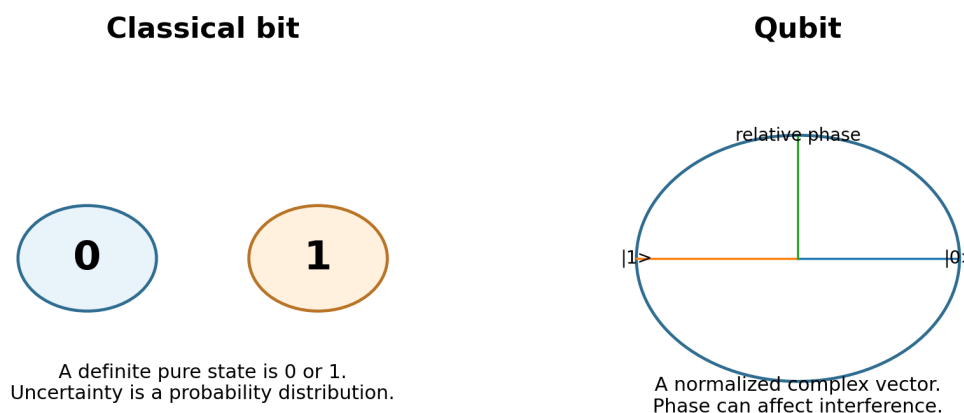


Figure 1. Classical uncertainty and quantum superposition are different mathematical objects.

A classical bit has two definite pure states, 0 and 1. A probability distribution can describe uncertainty about which state is present. A qubit also has two distinguished basis states, but its pure states are normalized complex linear combinations of those basis states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \alpha, \beta \in \mathbb{C}.$$

The complex phases of α and β can affect interference. This additional structure has no analogue in an ordinary classical probability distribution. A qubit is therefore not simply a bit that is partly 0 and partly 1.

Three layers of description

- Abstract state: a ray in a two-dimensional complex Hilbert space.
- Coordinate representation: a normalized two-component complex vector in a chosen basis.
- Physical implementation: two controllable energy levels, paths, polarizations, spin states, or other distinguishable quantum states.

2. Abstract state space of one qubit

The state space of one qubit is the two-dimensional complex vector space $\mathbb{C}^2$ equipped with an inner product. Mathematically, this is a Hilbert space.

$$H_{\text{qubit}} = \mathbb{C}^2.$$

A pure state is represented by a nonzero vector, with two further physical identifications:

- The vector is normalized so that total measurement probability is one.
- Vectors differing only by a global phase represent the same physical state.

$$|\psi\rangle \text{ and } e^{i\gamma}|\psi\rangle \text{ represent the same pure physical state.}$$

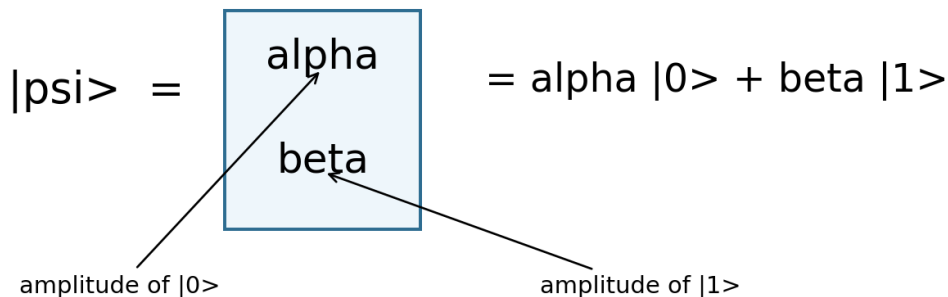
The term ray refers to all nonzero vectors that differ only by a nonzero scalar. After normalization, the remaining equivalence is multiplication by a unit complex phase.

Dimension is not the number of possible states

$\mathbb{C}^2$ has dimension two because two independent basis vectors are sufficient. Nevertheless, it contains infinitely many normalized vectors, and therefore infinitely many pure qubit states.

3. Computational basis and ket notation

Anatomy of a one-qubit state



α, β are complex numbers and $|\alpha|^2 + |\beta|^2 = 1$

Figure 2. The coordinates of a qubit state are probability amplitudes.

The standard computational basis is

$$|0\rangle = (1,0)^T, |1\rangle = (0,1)^T.$$

Every one-qubit vector has a unique expansion in this basis:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = (\alpha, \beta)^T.$$

The symbols $|0\rangle$ and $|1\rangle$ name vectors, not ordinary numbers. The labels are chosen because measurement in this basis produces classical outcomes 0 and 1.

Basis coordinates

The coefficient α is the coordinate along $|0\rangle$, obtained by the overlap $\langle 0|\psi\rangle$. Similarly $\beta = \langle 1|\psi\rangle$. Coordinates change when the basis changes, while the abstract state does not.

Orthogonality

$$\langle 0|0\rangle = 1, \langle 1|1\rangle = 1, \langle 0|1\rangle = 0.$$

4. Complex amplitudes and normalization

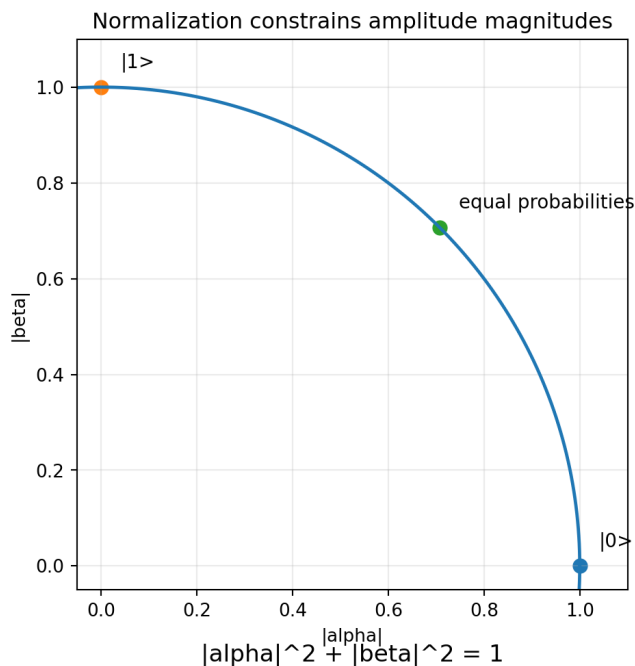


Figure 3. Amplitude magnitudes lie on a quarter-circle, while their phases supply additional degrees of freedom.

For a valid pure qubit state,

$$|\alpha|^2 + |\beta|^2 = 1.$$

This is equivalent to the vector normalization condition $\langle \psi | \psi \rangle = 1$. The amplitudes themselves need not be real or positive.

Example

$$|\psi\rangle = (\sqrt{3}/2)|0\rangle + (i/2)|1\rangle.$$

The squared magnitudes are 3/4 and 1/4, so the state is normalized.

Normalizing an arbitrary nonzero vector

Given $|v\rangle = (a, b)^T$, compute

$$N = \sqrt{|a|^2 + |b|^2}, \quad |\psi\rangle = |v\rangle / N.$$

The zero vector cannot be normalized and does not represent a physical state.

5. Born rule and measurement probabilities

Born rule: square the amplitude magnitudes

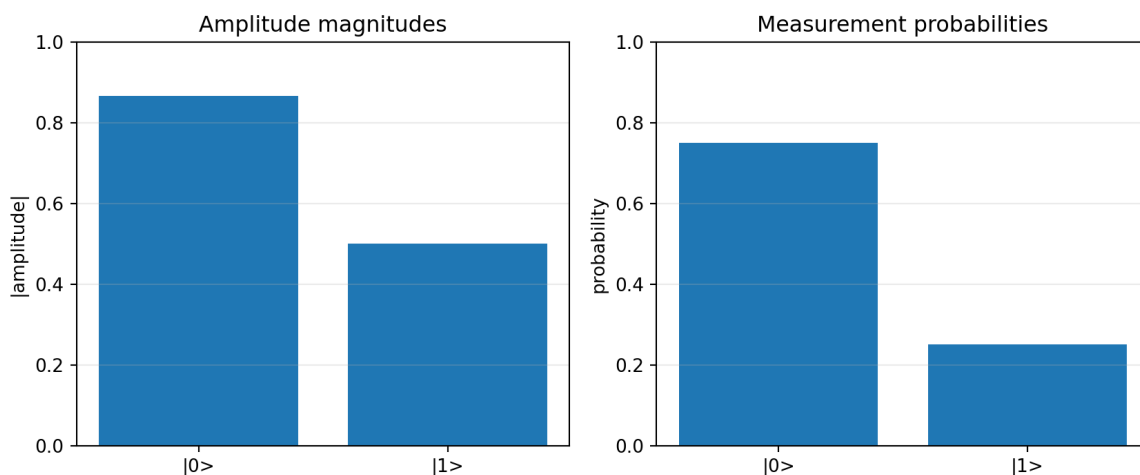


Figure 4. Measurement probability is the squared magnitude of an amplitude.

For computational-basis measurement of

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

the Born rule gives

$$P(0) = |\alpha|^2, P(1) = |\beta|^2.$$

More generally, for a normalized measurement outcome $|\phi\rangle$,

$$P(\phi) = |\langle\phi|\psi\rangle|^2.$$

Amplitude versus probability

- An amplitude may be negative or complex.
- A probability is real and non-negative.
- Amplitudes for indistinguishable alternatives add before the squared magnitude is taken.
- Probabilities of mutually exclusive measurement outcomes sum to one.

Repeated shots

A single measurement yields one classical outcome. Repeating the same preparation and measurement many times estimates the theoretical probability distribution. Finite counts fluctuate statistically around the Born-rule probabilities.

6. Global and relative phase

Only relative phase changes observable interference

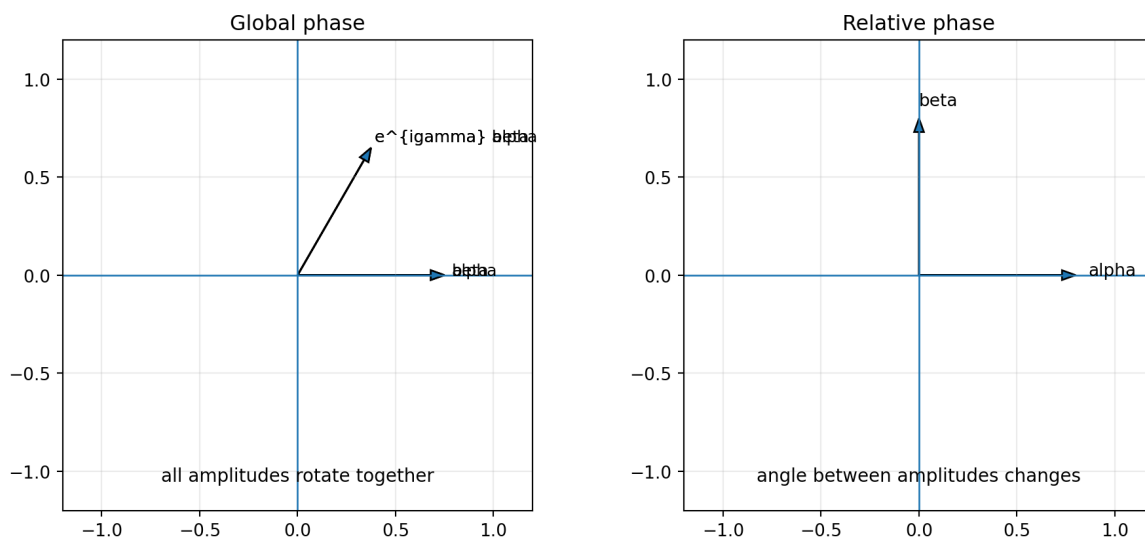


Figure 5. A common rotation of all amplitudes is global; their phase difference is relative.

Write the amplitudes in polar form:

$$\alpha = r_0 e^{i\theta_0}, \beta = r_1 e^{i\theta_1}.$$

Factor out $e^{i\theta_0}$:

$$|\psi\rangle = e^{i\theta_0} (r_0|0\rangle + r_1 e^{i(\theta_1 - \theta_0)}|1\rangle).$$

The outer factor is global phase and has no effect on isolated-state measurement probabilities or future unitary evolution. The physically relevant phase inside the superposition is the relative phase

$$\phi = \theta_1 - \theta_0.$$

Example

The states $(|0\rangle + |1\rangle)/\sqrt{2}$ and $i(|0\rangle + |1\rangle)/\sqrt{2}$ are physically equivalent. The states $(|0\rangle + |1\rangle)/\sqrt{2}$ and $(|0\rangle + i|1\rangle)/\sqrt{2}$ are not, because their relative phases differ.

7. Common one-qubit basis states

Two common orthonormal bases for one qubit

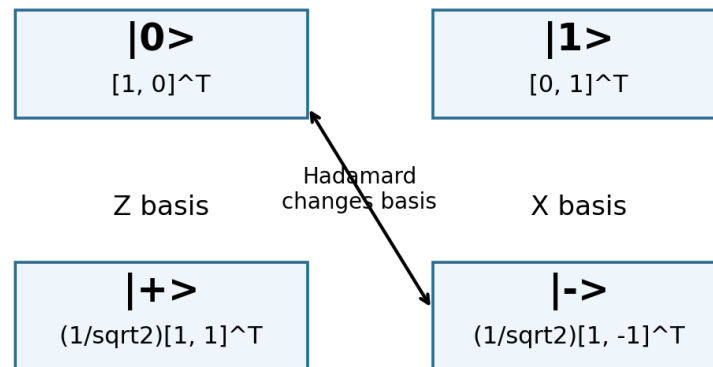


Figure 6. The Z and X bases describe the same two-dimensional space using different axes.

State	Vector	Bloch direction	Definite observable
$ 0\rangle$	$(1,0)^T$	+z	$Z=+1$
$ 1\rangle$	$(0,1)^T$	-z	$Z=-1$
$ +\rangle$	$(1,1)^T/\sqrt{2}$	+x	$X=+1$
$ -\rangle$	$(1,-1)^T/\sqrt{2}$	-x	$X=-1$
$ +i\rangle$	$(1,i)^T/\sqrt{2}$	+y	$Y=+1$
$ -i\rangle$	$(1,-i)^T/\sqrt{2}$	-y	$Y=-1$

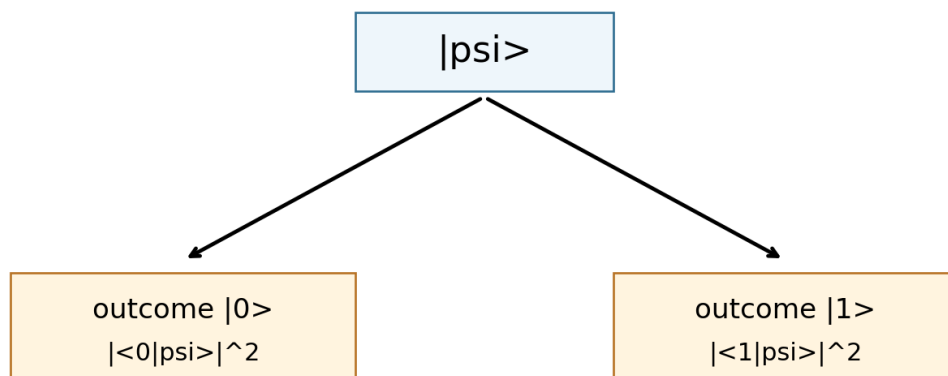
Each pair forms an orthonormal basis. The same physical state may have different coordinates in these bases. For example:

$$|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}.$$

Therefore a Z-basis eigenstate produces random outcomes in the X basis.

8. Measurement as projection

Measurement branches according to squared overlaps



After an ideal projective outcome, the state becomes the corresponding basis state.

Figure 7. A projective measurement converts overlaps into probabilities and selects an outcome subspace.

For computational-basis measurement, define projectors

$$P_0 = |0\rangle\langle 0|, P_1 = |1\rangle\langle 1|.$$

They satisfy

$$P_0 + P_1 = I, P_j^2 = P_j, P_0 P_1 = 0.$$

The outcome probabilities are

$$p_j = \langle \psi | P_j | \psi \rangle.$$

If outcome j is obtained, the ideal post-measurement state is

$$P_j |\psi\rangle / \sqrt{p_j}.$$

For a one-dimensional basis projector, this reduces to the corresponding basis state up to phase.

Measurement is not ordinary unitary evolution

Conditioning on a particular outcome is not reversible. Before reading the result, the overall measurement process can be modeled using a larger quantum system and unitary interaction, but the observed branch is described by projection and renormalization.

9. Measuring in X and Y bases

A basis measurement can be implemented by rotating into the Z basis

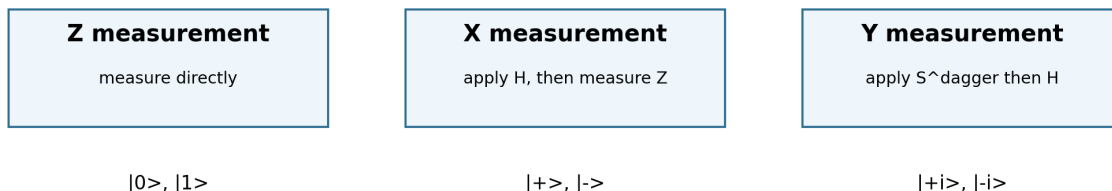


Figure 8. Hardware that measures Z can measure other bases after a suitable basis rotation.

X-basis probabilities

$$P(+)=|\langle +|\psi\rangle|^2, P(-)=|\langle -|\psi\rangle|^2.$$

Because $H|+\rangle=|0\rangle$ and $H|-\rangle=|1\rangle$, applying H before a Z measurement implements an X measurement.

Y-basis probabilities

$$P(+i)=|\langle +i|\psi\rangle|^2, P(-i)=|\langle -i|\psi\rangle|^2.$$

A standard implementation applies $S^\dagger$ and then H before measuring Z.

Example: phase becomes visible

The states $|+\rangle$ and $|-\rangle$ both give 50/50 probabilities in the Z basis. In the X basis, $|+\rangle$ gives the + outcome with certainty, while $|-\rangle$ gives the - outcome with certainty. Thus basis choice reveals relative phase.

10. Bloch sphere geometry

Pure one-qubit states on the Bloch sphere

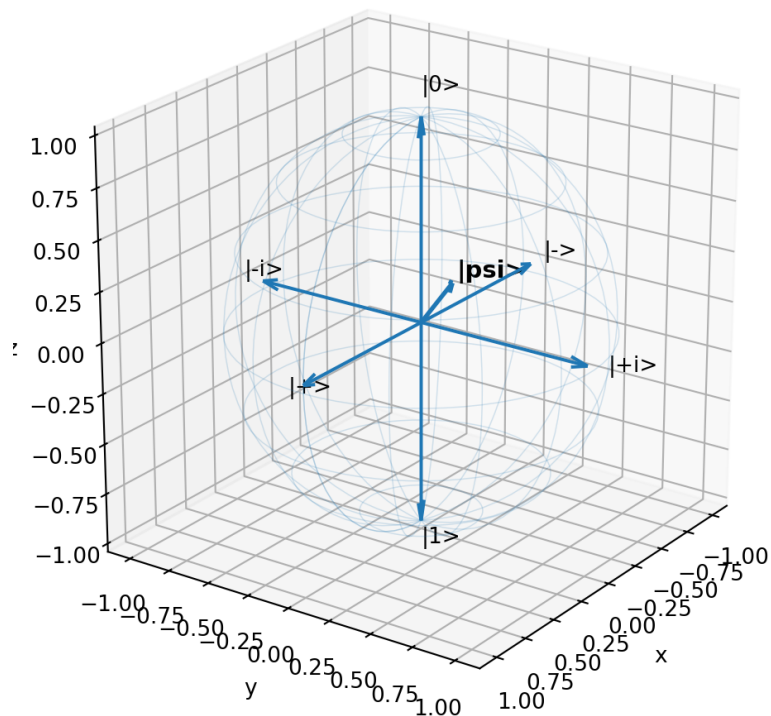


Figure 9. Every pure one-qubit state corresponds to a point on the Bloch sphere.

After removing normalization and global phase, a pure qubit has two real degrees of freedom. These can be represented by a point on a sphere.

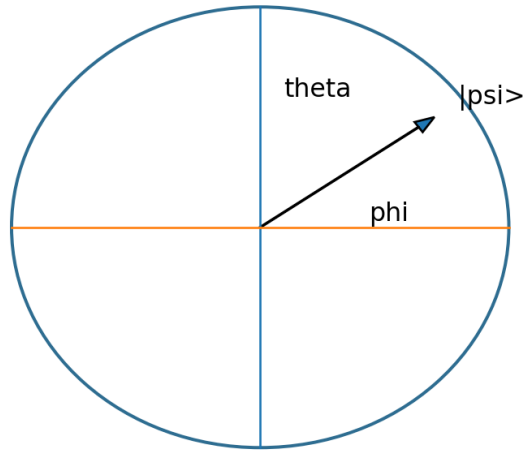
Important axes

- North and south poles: $|0\rangle$ and $|1\rangle$.
- Positive and negative x-axis: $|+\rangle$ and $|-\rangle$.
- Positive and negative y-axis: $|+i\rangle$ and $|-i\rangle$.

The Bloch sphere represents pure one-qubit states only. It is not a literal physical sphere inside the qubit. It is a geometric representation of state parameters and measurement expectations.

11. Bloch parameterization

Bloch-sphere parameterization



$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$$

Figure 10. Polar angle θ determines the population balance; azimuth ϕ is relative phase.

Every pure one-qubit state can be written, after removing global phase, as

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle,$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$.

Measurement probabilities

$$P(0) = \cos^2(\theta/2), \quad P(1) = \sin^2(\theta/2).$$

The azimuthal angle ϕ does not affect direct Z-basis probabilities, but it changes X- and Y-basis statistics.

Bloch vector

$$\mathbf{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

For a pure state, $|\mathbf{r}|=1$.

12. Pure states and global-phase equivalence

A normalized two-component complex vector appears to contain four real parameters. The physical pure state has only two:

- 1 Normalization removes one real degree of freedom.
- 2 Global-phase equivalence removes another real degree of freedom.

This leaves θ and ϕ , the coordinates on the Bloch sphere.

Testing global equivalence

Two normalized vectors $|\psi\rangle$ and $|\phi\rangle$ represent the same pure state exactly when there exists a unit-modulus scalar $e^{i\gamma}$ such that

$$|\phi\rangle = e^{i\gamma} |\psi\rangle.$$

Equivalently, their overlap has unit magnitude:

$$|\langle\phi|\psi\rangle| = 1.$$

Example

$$(i, -1)^T / \sqrt{2} = i(1, i)^T / \sqrt{2}.$$

These vectors are algebraically different but physically identical as isolated pure states.

13. Single-qubit gates as transformations

A closed-system one-qubit gate is a 2×2 unitary matrix U acting as

$$|\psi_{\text{out}}\rangle = U|\psi_{\text{in}}\rangle.$$

Unitarity ensures

$$U^\dagger U = I,$$

which preserves norms, inner products, orthogonality, and total probability. The matrix columns are $U|0\rangle$ and $U|1\rangle$, so the action on the computational basis determines the action on every superposition.

Composition order

$$U_3 U_2 U_1 |\psi\rangle \text{ means } U_1 \text{ acts first.}$$

Reversibility

Every unitary gate has inverse $U^{-1} = U^\dagger$. Measurement and uncontrolled noise are not generally represented by unitary operations on the qubit alone.

Global phase of a gate

U and $e^{i\gamma}U$ produce output states differing only by global phase for every input, so they describe the same isolated-state action. Relative phase between matrix components still matters.

14. Pauli, Hadamard, and phase gates

Gate	Matrix	Main action
X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	swaps $ 0\rangle$ and $ 1\rangle$
Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	bit flip with phases
Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	adds π relative phase to $ 1\rangle$
H	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	changes between Z and X bases
S	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	adds $\pi/2$ phase to $ 1\rangle$
T	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$	adds $\pi/4$ phase to $ 1\rangle$

Examples

$$X(\alpha, \beta)^T = (\beta, \alpha)^T.$$

$$Z(\alpha, \beta)^T = (\alpha, -\beta)^T.$$

$$H(\alpha, \beta)^T = ((\alpha + \beta)/\sqrt{2}, (\alpha - \beta)/\sqrt{2})^T.$$

The Hadamard formula displays interference explicitly: amplitudes add in one output and subtract in the other.

15. Rotations on the Bloch sphere

Single-qubit gates act as rotations or reflections of coordinates

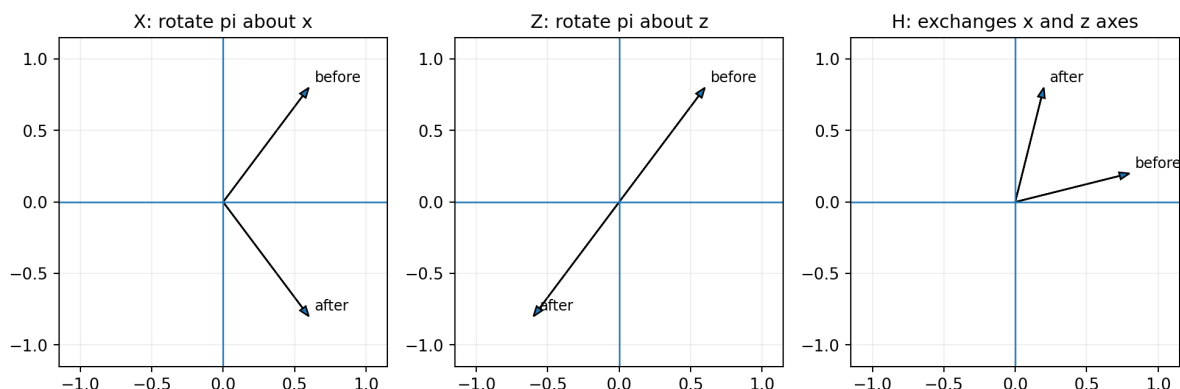


Figure 11. Unitary single-qubit gates transform the Bloch vector while preserving its length.

Ignoring global phase, every single-qubit unitary corresponds to a three-dimensional rotation of the Bloch sphere.

Rotation operators

$$R_x(\theta) = \exp(-i \theta X/2), R_y(\theta) = \exp(-i \theta Y/2), R_z(\theta) = \exp(-i \theta Z/2).$$

Pauli gates are π rotations about their respective axes, up to global phase. The phase gate S is a $\pi/2$ rotation about z up to global phase, and T is a $\pi/4$ z rotation up to global phase.

Why half-angles appear

The state vector uses $\cos(\theta/2)$ and $\sin(\theta/2)$, while the Bloch vector rotates by θ . This double-cover relationship is a characteristic feature of spin-1/2 systems.

16. Expectation values of X, Y, and Z

For a state with Bloch vector $r=(r_x, r_y, r_z)$,

$$r_x=\langle X \rangle, \quad r_y=\langle Y \rangle, \quad r_z=\langle Z \rangle.$$

The expectation values are

$$\langle X \rangle = \langle \psi | X | \psi \rangle, \quad \langle Y \rangle = \langle \psi | Y | \psi \rangle, \quad \langle Z \rangle = \langle \psi | Z | \psi \rangle.$$

If $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then

$$\langle X \rangle = 2 \operatorname{Re}(\alpha^* \beta),$$

$$\langle Y \rangle = 2 \operatorname{Im}(\alpha^* \beta),$$

$$\langle Z \rangle = |\alpha|^2 - |\beta|^2.$$

These three real numbers determine a one-qubit density matrix:

$$\rho = (I + r_x X + r_y Y + r_z Z)/2.$$

Operational meaning

Each Pauli expectation is estimated by repeated measurement in the corresponding eigenbasis and averaging outcomes coded as +1 and -1.

17. Density matrices and mixed states

A pure state $|\psi\rangle$ can be represented by the density matrix

$$\rho = |\psi\rangle\langle\psi|.$$

For $|\psi\rangle = (\alpha, \beta)^T$,

$$\rho = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}.$$

A statistical mixture of states $|\psi_k\rangle$ with classical probabilities p_k is

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|.$$

Every valid density matrix satisfies:

- ρ is Hermitian.
- ρ is positive semidefinite.
- $\text{Tr}(\rho) = 1$.

Measurement probability becomes

$$p_j = \text{Tr}(P_j \rho).$$

Density matrices describe both quantum coherence and classical uncertainty in one formalism.

18. Pure versus mixed states

The Bloch sphere becomes a Bloch ball for density matrices

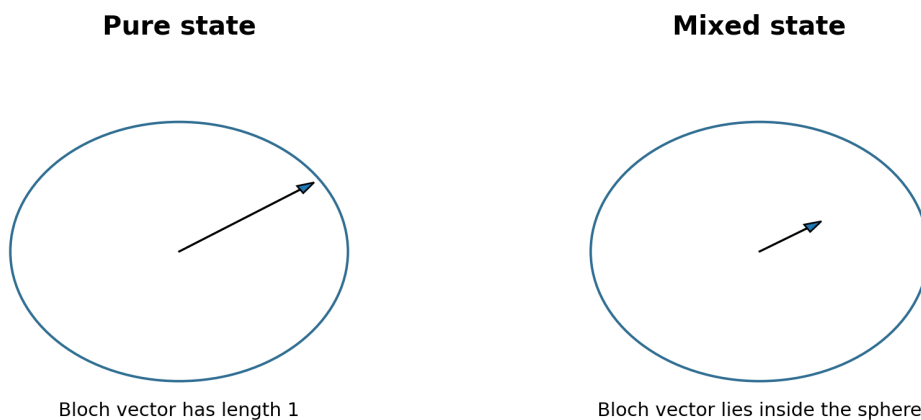


Figure 12. Pure states lie on the Bloch-sphere surface; mixed states lie inside.

A density matrix is pure exactly when

$$\rho^2 = \rho, \text{ equivalently } \text{Tr}(\rho^2) = 1.$$

A mixed state has

$$\text{Tr}(\rho^2) < 1.$$

Example: coherent superposition versus incoherent mixture

The pure state $|+\rangle$ has

$$\rho_+ = (1/2)[[1, 1], [1, 1]].$$

The equal classical mixture of $|0\rangle$ and $|1\rangle$ has

$$\rho_{\text{mix}} = (1/2)[[1, 0], [0, 1]] = I/2.$$

Both produce 50/50 outcomes in the Z basis. In the X basis, $|+\rangle$ gives + with certainty, while $I/2$ remains 50/50. The off-diagonal entries encode coherence in the chosen basis.

19. State preparation and tomography

State preparation means applying controlled operations so that a reference state, usually $|0\rangle$, becomes a desired state.

$$|\psi\rangle = R_z(\phi) R_y(\theta)|0\rangle \text{ up to global phase.}$$

A single computational-basis measurement cannot determine an unknown qubit state because it reveals only the Z-basis populations. State tomography estimates X, Y, and Z expectation values from separate measurement settings.

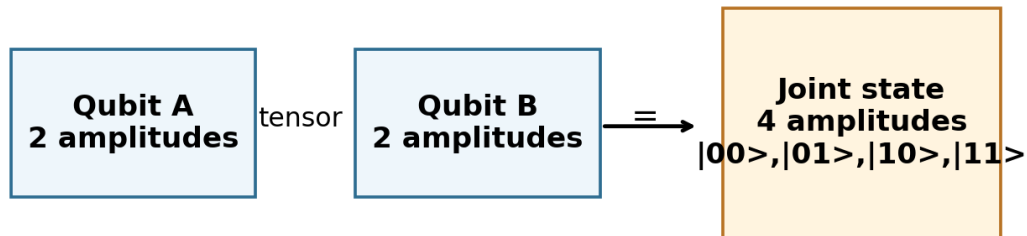
One-qubit tomography workflow

- 1 Prepare the same state many times.
- 2 Measure a subset directly in Z.
- 3 Apply H and measure another subset to estimate X.
- 4 Apply $S^\dagger$ then H and measure another subset to estimate Y.
- 5 Reconstruct $\rho = (I + \langle X \rangle X + \langle Y \rangle Y + \langle Z \rangle Z)/2$.

Tomography requires many copies of identically prepared states. A single unknown qubit cannot be fully identified from one measurement.

20. Combining qubits with tensor products

Combining qubits multiplies dimensions



$$\dim(H_A \text{ tensor } H_B) = \dim(H_A) \dim(H_B) = 2 \times 2 = 4$$

Figure 13. The joint state space is the tensor product of subsystem spaces.

For two qubits A and B,

$$H_{AB} = H_A \text{ tensor } H_B, \dim(H_{AB}) = 2 \times 2 = 4.$$

The computational basis is

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

If

$$|\psi\rangle_A = \alpha|0\rangle + \beta|1\rangle, |\phi\rangle_B = \gamma|0\rangle + \delta|1\rangle,$$

then their product state is

$$|\psi\rangle \text{ tensor } |\phi\rangle = \alpha \gamma |00\rangle + \alpha \delta |01\rangle + \beta \gamma |10\rangle + \beta \delta |11\rangle.$$

For n qubits, the state vector has 2^n amplitudes. This exponential growth is central to both quantum information and the difficulty of classical simulation.

21. Product states and entanglement

Not every two-qubit state is a product of two one-qubit states

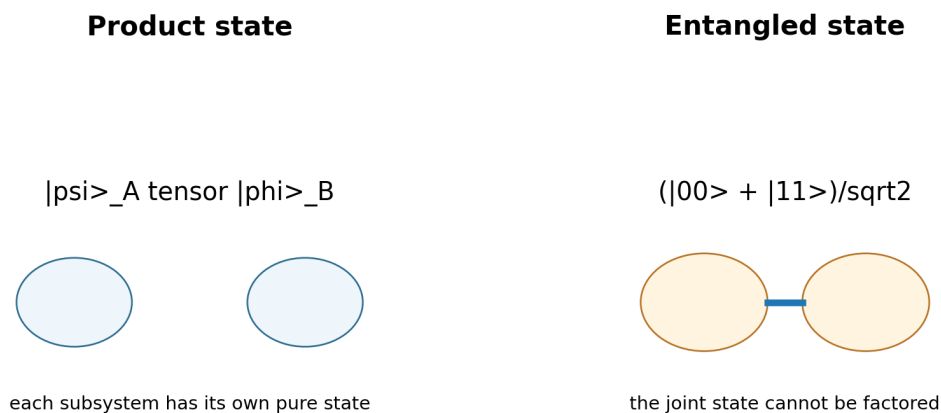


Figure 14. Entangled states cannot be decomposed into independent subsystem state vectors.

A two-qubit pure state is a product state if it can be written as $|\psi\rangle_A \text{ tensor } |\phi\rangle_B$. Otherwise it is entangled.

Bell-state example

$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}.$$

Suppose it were a product $(a|0\rangle + b|1\rangle) \text{ tensor } (c|0\rangle + d|1\rangle)$. Matching coefficients would require ac and bd nonzero while $ad=bc=0$, which is impossible. Therefore the state is entangled.

Subsystem description

The joint Bell state is pure, but either qubit alone is described by the maximally mixed state $I/2$. Entanglement therefore creates correlations that cannot be explained by assigning independent pure states to the subsystems.

22. No-cloning and measurement disturbance

There is no unitary operation that copies every unknown quantum state:

$$|\psi\rangle|0\rangle \rightarrow |\psi\rangle|\psi\rangle \text{ for all } |\psi\rangle \text{ is impossible.}$$

Linearity proof sketch

A copier that works on $|0\rangle$ and $|1\rangle$ would map

$$|0\rangle|0\rangle \rightarrow |0\rangle|0\rangle, |1\rangle|0\rangle \rightarrow |1\rangle|1\rangle.$$

By linearity, it would map $|+\rangle|0\rangle$ to $(|00\rangle+|11\rangle)/\sqrt{2}$, but perfect copying would require $|+\rangle|+\rangle=(|00\rangle+|01\rangle+|10\rangle+|11\rangle)/2$. These differ.

Measurement disturbance

Measuring an unknown state generally changes it. A Z measurement of $|+\rangle$ produces 0 or 1 and leaves the system in $|0\rangle$ or $|1\rangle$, destroying the original X-basis coherence. This is not merely imperfect equipment; it follows from the structure of projective measurement.

Cryptographic relevance

No-cloning and measurement disturbance contribute to the security intuition of quantum key distribution: an eavesdropper cannot perfectly copy unknown nonorthogonal signal states and measure them later without risk of detectable disturbance.

23. Physical implementations of qubits

Different physical systems can implement the same abstract qubit

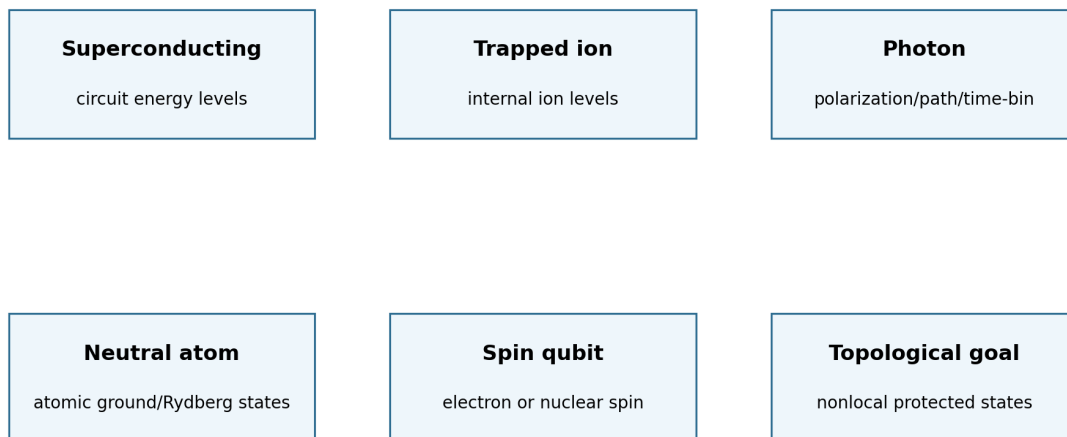


Figure 15. The same abstract qubit can be encoded in many physical platforms.

A physical qubit requires:

- Two well-defined logical states that can be initialized.
- Controllable transformations approximating desired unitary gates.
- A measurement process that distinguishes relevant basis states.
- Sufficient isolation to preserve coherence, together with controlled coupling for gates.

Logical versus physical qubits

A physical qubit is one imperfect hardware degree of freedom. A logical qubit is encoded across many physical qubits using quantum error correction so that noise can be detected and corrected without directly measuring the protected logical amplitudes.

Platform independence

The vector, matrix, and measurement mathematics is largely independent of implementation. Hardware differences enter through available gates, connectivity, timing, noise channels, control systems, and readout models.

24. Noise, decoherence, and fidelity

Real qubits interact with uncontrolled environments and imperfect controls. The state may therefore evolve through a quantum channel rather than a unitary gate.

Relaxation and dephasing

- Energy relaxation changes population, commonly driving $|1\rangle$ toward $|0\rangle$; its characteristic timescale is often denoted T_1 .
- Dephasing destroys relative phase coherence without necessarily changing Z-basis populations; its timescale is related to T_2 .

Simple noise channels

- Bit-flip channel: applies X with some probability.
- Phase-flip channel: applies Z with some probability.
- Depolarizing channel: replaces part of the state with maximally mixed noise.
- Amplitude damping: models energy loss.

Fidelity

For a target pure state $|\psi\rangle$ and actual state ρ , state fidelity is

$$F = \langle \psi | \rho | \psi \rangle.$$

It equals one for perfect preparation and decreases as the actual state departs from the target. Error mitigation, calibration, and quantum error correction address different aspects of these imperfections.

25. Worked examples

Example 1: normalize a state

Normalize $(1+i, 2)^T$. The squared norm is $2+4=6$, so

$$|\psi\rangle = (1/\sqrt{6})(1+i, 2)^T.$$

Example 2: Z-basis probabilities

For $|\psi\rangle = (\sqrt{3}/2)|0\rangle + (i/2)|1\rangle$,

$$P(0)=3/4, P(1)=1/4.$$

Example 3: X-basis probability

For the same state,

$$\langle +|\psi\rangle = (1/\sqrt{2})(\sqrt{3}/2 + i/2) = (\sqrt{3}+i)/(2\sqrt{2}),$$

$$P(+)=|\sqrt{3}+i|^2/8=4/8=1/2.$$

Example 4: Bloch coordinates

For $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$, the Bloch vector is

$$(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

Example 5: density matrix

For $|+\rangle = (1, i)^T/\sqrt{2}$,

$$\rho = (1/2)[[1, -i], [i, 1]].$$

26. Common mistakes

Mistake	Correction
A qubit is simultaneously a classical 0 and 1	A pure qubit is a complex state vector; measurement outcomes are classical.
Amplitude equals probability	Probability is squared magnitude of amplitude.
All phases are unobservable	Only common global phase is unobservable; relative phase affects interference.
The Bloch sphere shows all multi-qubit states	It represents one-qubit density matrices; multi-qubit spaces are much larger.
Measurement merely reveals a pre-existing basis value	For a general state, measurement is probabilistic and changes the state.
A mixed state is a coherent superposition	A mixture represents classical uncertainty; coherence is stored in off-diagonal terms.
CNOT clones a qubit	It copies basis values but entangles an arbitrary superposition.
One measurement determines an unknown state	Tomography requires many identically prepared copies and multiple bases.

A reliable habit is to distinguish four objects: state vector, density matrix, measurement basis, and classical outcome distribution. Each answers a different question.

27. Exercises

- 1 Normalize the vector $(2, 1-i)^T$.
- 2 For $|\psi\rangle = (1/2)|0\rangle + (\sqrt{3}/2)|1\rangle$, find Z-basis probabilities.
- 3 For $|\psi\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$, find $P(+)$ and $P(-)$.
- 4 Determine whether $(1, i)^T/\sqrt{2}$ and $(i, -1)^T/\sqrt{2}$ differ only by global phase.
- 5 Write $|1\rangle$ in the X basis.
- 6 Find θ and ϕ for $|\psi\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$.
- 7 Compute $\langle X \rangle$, $\langle Y \rangle$, and $\langle Z \rangle$ for $|+\rangle$.
- 8 Apply H to $|-\rangle$.
- 9 Apply Z to $|+\rangle$.
- 10 Calculate the density matrix of $|-\rangle$.
- 11 Compare the density matrix of $|+\rangle$ with the equal mixture of $|0\rangle$ and $|1\rangle$.
- 12 Show that the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$ is normalized.
- 13 Determine whether $(|00\rangle + |01\rangle + |10\rangle + |11\rangle)/2$ is entangled.
- 14 Explain why a Z measurement of $|+\rangle$ destroys its definite X value.
- 15 Give a basis-rotation circuit for measuring Y.
- 16 If $\langle Z \rangle = 0.6$, what is the difference $P(0) - P(1)$?
- 17 A target pure state $|\psi\rangle$ has measured fidelity 0.92 with ρ . Interpret the number.
- 18 Explain why an arbitrary qubit cannot be copied even though a classical basis bit can be copied with CNOT.
- 19 How many amplitudes describe a general five-qubit pure state?
- 20 List the mathematical differences among a pure state vector, a density matrix, and a measurement outcome.

28. Selected solutions

- 1 The squared norm is $4+|1-i|^2=6$, so the normalized state is $(2,1-i)^T/\sqrt{6}$.
- 2 $P(0)=1/4$ and $P(1)=3/4$.
- 3 $\langle +|\psi\rangle=(1+i)/2$ and $\langle -|\psi\rangle=(1-i)/2$, so both probabilities are $1/2$.
- 4 Yes. Multiplying $(1,i)^T/\sqrt{2}$ by i gives $(i,-1)^T/\sqrt{2}$.
- 5 $|1\rangle=(|+\rangle-|-\rangle)/\sqrt{2}$.
- 6 $\theta=\pi/2$ and $\phi=\pi/2$.
- 7 $|+i\rangle$ is the $+1$ Y eigenstate, so $\langle X\rangle=0$, $\langle Y\rangle=1$, $\langle Z\rangle=0$.
- 8 $H|-\rangle=|1\rangle$.
- 9 $Z|+\rangle=|-\rangle$.
- 10 $\rho_-=(1/2)[[1,-1],[-1,1]]$.
- 11 $\rho_+=(1/2)[[1,1],[1,1]]$, while the mixture is $I/2$; only the pure state has off-diagonal coherence.
- 12 The two nonzero amplitudes have squared magnitudes $1/2$ and $1/2$, summing to one.
- 13 It factors as $|+\rangle$ tensor $|+\rangle$, so it is not entangled.
- 14 The outcome projects the state onto $|0\rangle$ or $|1\rangle$; either resulting state has random X measurement and no longer equals $|+\rangle$.
- 15 Apply $S^\dagger$, then H , then perform a computational-basis measurement.
- 16 $\langle Z\rangle=P(0)-P(1)=0.6$.
- 17 The actual state has 92 percent overlap probability with the target under the fidelity definition $F=\langle\psi|\rho|\psi\rangle$.
- 18 CNOT copies $|0\rangle$ and $|1\rangle$ basis values, but linearity maps a superposition to an entangled state rather than two independent copies.
- 19 $2^5=32$ complex amplitudes, constrained by normalization and global phase.
- 20 A pure vector is a normalized amplitude representation up to global phase; a density matrix represents pure or mixed states; a measurement outcome is a classical label sampled according to the measurement probabilities.

29. Reference sheet

Concept	Formula or fact
Pure qubit	$ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$
Normalization	$ \alpha ^2 + \beta ^2 = 1$
Born rule	$P(\phi) = \langle\phi \psi\rangle ^2$
Global phase	$ \psi\rangle$ equivalent to $e^{i\gamma} \psi\rangle$
Bloch form	$\cos(\theta/2) 0\rangle + e^{i\phi}\sin(\theta/2) 1\rangle$
Bloch vector	$(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$
Z probabilities	$\cos^2(\theta/2), \sin^2(\theta/2)$
Density matrix	$\rho = \psi\rangle\langle\psi $ or $\sum p_k \psi_k\rangle\langle\psi_k $
Purity	$\text{Tr}(\rho^2) = 1$ for pure; < 1 for mixed
Pauli expectations	$r_x = \langle X \rangle, r_y = \langle Y \rangle, r_z = \langle Z \rangle$
Unitary evolution	$ \psi_{\text{out}}\rangle = U \psi_{\text{in}}\rangle, U^\dagger U = I$
Projective probability	$p_j = \text{Tr}(P_j \rho)$
Two-qubit basis	$ 00\rangle, 01\rangle, 10\rangle, 11\rangle$
n-qubit dimension	2^n
Pure-state fidelity	$F = \langle\psi \rho \psi\rangle$

Essential conceptual map

- Complex amplitudes encode both probability magnitude and phase.
- Measurement basis determines which state components become classical outcomes.
- The Bloch sphere visualizes all one-qubit density matrices: pure on the surface, mixed inside.
- Unitary gates preserve the quantum information of a closed system.
- Tensor products create multi-qubit spaces; entanglement appears when the joint vector cannot be factored.

The next topic is Quantum Measurement: measurement postulates, observables, projective measurements, POVMs, state update, expectation values, and experimental statistics.